

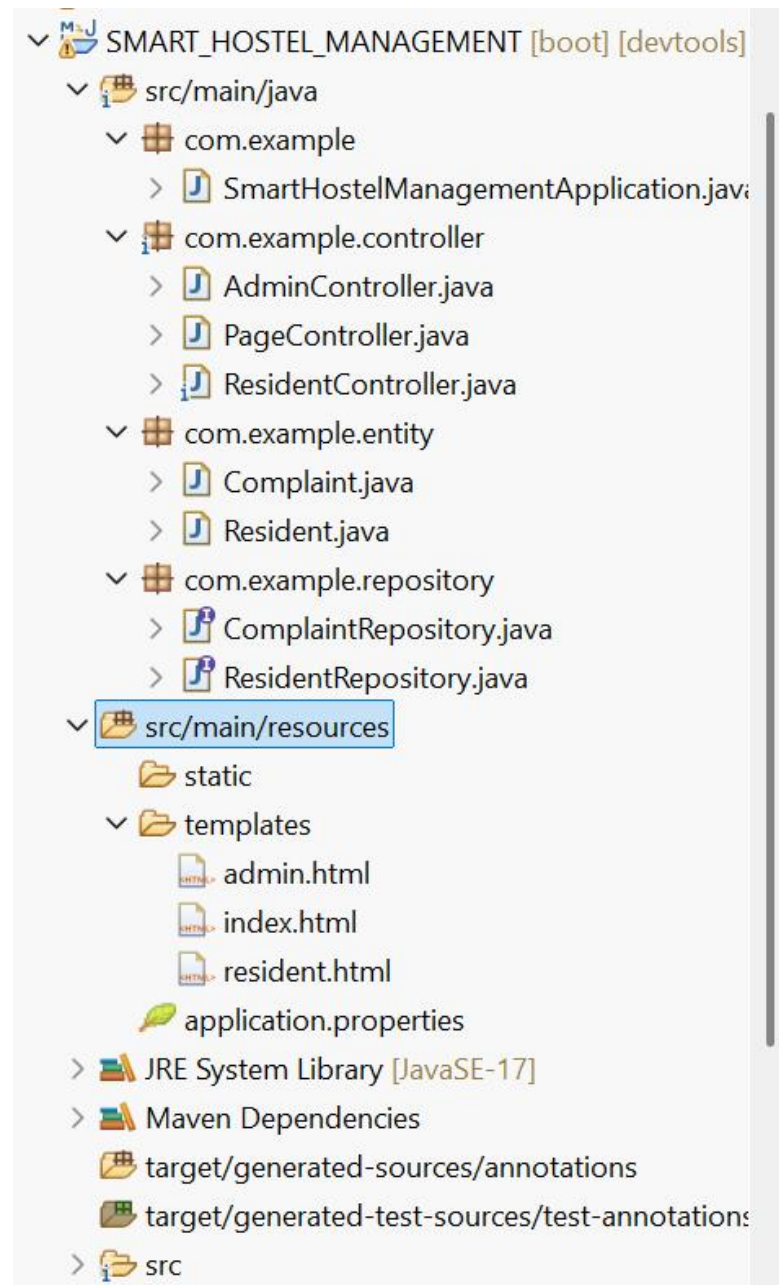
TEAM NO : 16

SECTION : 8

FSAD PROJECT CODES:

PROJECT: SMART HOSTEL MANAGEMENT

1. FOLDER/ARCHITECTURE:



2. SmartHostelManagementApplication.java(Com.example)

```
package com.example;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class SmartHostelManagementApplication {

    public static void main(String[] args) {
        SpringApplication.run(SmartHostelManagementApplication.class, args);
    }

}
```

3. 1.AdminController.java(Com.example.Controller)

```
package com.example.controller;

import com.example.entity.*;
import com.example.repository.*;
import org.springframework.web.bind.annotation.*;

import java.util.List;

@RestController
@RequestMapping("/admin")
@CrossOrigin
public class AdminController {

    private final ResidentRepository residentRepo;
    private final ComplaintRepository complaintRepo;

    public AdminController(ResidentRepository r, ComplaintRepository c){
        this.residentRepo = r;
        this.complaintRepo = c;
    }

    @PostMapping("/add")
    public Resident add(@RequestBody Resident r){
        return residentRepo.save(r);
    }

    @GetMapping("/residents")
    public List<Resident> getResidents(){
        return residentRepo.findAll();
    }

    @GetMapping("/complaints")
    public List<Complaint> getComplaints(){
        return complaintRepo.findAll();
    }

    @PutMapping("/complaint/{id}")
    public Complaint reply(@PathVariable Long id, @RequestBody Complaint c){
```

```

        Complaint old = complaintRepo.findById(id).orElseThrow();
        old.setStatus("Resolved");
        old.setResponse(c.getResponse());
        return complaintRepo.save(old);
    }
}

```

2. PageController.java(Com.example.Controller)

```

package com.example.controller;

import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.GetMapping;

@Controller
public class PageController {

    @GetMapping("/")
    public String home(){
        return "index";
    }

    @GetMapping("/admin")
    public String admin(){
        return "admin";
    }

    @GetMapping("/resident")
    public String resident(){
        return "resident";
    }
}

```

3. ResidentController.java(Com.example.Controller)

```

package com.example.controller;

import com.example.entity.*;
import com.example.repository.*;
import org.springframework.web.bind.annotation.*;
import org.springframework.web.multipart.MultipartFile;

import java.io.File;

@RestController
@RequestMapping("/resident")
@CrossOrigin
public class ResidentController {

    private final ResidentRepository residentRepo;
    private final ComplaintRepository complaintRepo;

    public ResidentController(ResidentRepository r, ComplaintRepository c){
        this.residentRepo = r;
        this.complaintRepo = c;
    }
}

```

```

    @GetMapping("/{id}")
    public Resident get(@PathVariable Long id){
        return residentRepo.findById(id).orElseThrow();
    }

    @PostMapping("/complaint")
    public Complaint add(@RequestBody Complaint c){
        return complaintRepo.save(c);
    }

    @PostMapping("/upload/{id}")
    public String upload(@PathVariable Long id,
        @RequestParam("file") MultipartFile file) throws Exception {

        String path = "src/main/resources/static/uploads/" + file.getOriginalFilename();
        file.transferTo(new File(path));

        Resident r = residentRepo.findById(id).orElseThrow();
        r.setPaymentProof(file.getOriginalFilename());
        residentRepo.save(r);

        return "Uploaded";
    }
}

```

3 1. Complaint.java(Com.example.entity)

```

package com.example.entity;

import jakarta.persistence.*;

@Entity
public class Complaint {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String residentName;
    private String issue;
    private String status = "Pending";
    private String response;

    public Long getId(){ return id; }
    public void setId(Long id){ this.id = id; }

    public String getResidentName(){ return residentName; }
    public void setResidentName(String residentName){ this.residentName = residentName; }

    public String getIssue(){ return issue; }
    public void setIssue(String issue){ this.issue = issue; }

    public String getStatus(){ return status; }
    public void setStatus(String status){ this.status = status; }
}

```

```

public String getResponse(){ return response; }
public void setResponse(String response){ this.response = response; }
}

```

2. Resident.java(Com.example.entity)

```

package com.example.entity;

import jakarta.persistence.*;

@Entity
public class Resident {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;
    private String phone;
    private String address;
    private int roomNumber;
    private double rentDue;
    private String paymentProof;

    // getters & setters
    public Long getId(){ return id; }
    public void setId(Long id){ this.id = id; }

    public String getName(){ return name; }
    public void setName(String name){ this.name = name; }

    public String getPhone(){ return phone; }
    public void setPhone(String phone){ this.phone = phone; }

    public String getAddress(){ return address; }
    public void setAddress(String address){ this.address = address; }

    public int getRoomNumber(){ return roomNumber; }
    public void setRoomNumber(int roomNumber){ this.roomNumber = roomNumber; }

    public double getRentDue(){ return rentDue; }
    public void setRentDue(double rentDue){ this.rentDue = rentDue; }

    public String getPaymentProof(){ return paymentProof; }
    public void setPaymentProof(String paymentProof){ this.paymentProof =
paymentProof; }
}

```

4 1. ComplaintRepository.java(Com.example.Repository)

```

package com.example.repository;

import com.example.entity.Complaint;
import org.springframework.data.jpa.repository.JpaRepository;

```

```
public interface ComplaintRepository extends JpaRepository<Complaint, Long> {  
}
```

2 ResidentRepository.java(Com.example.Repository)

```
package com.example.repository;  
  
import com.example.entity.Resident;  
import org.springframework.data.jpa.repository.JpaRepository;  
  
public interface ResidentRepository extends JpaRepository<Resident, Long> {  
}
```

4. Templates:

Admin.html

```
<!DOCTYPE html>  
<html>  
<head>  
<title>Admin Dashboard</title>  
  
<link  
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;700  
&display=swap" rel="stylesheet">  
  
<style>  
*{margin:0;padding:0;box-sizing:border-box;font-family:'Poppins',sans-  
serif;}  
  
/* ✓ ADDED */  
html{  
scroll-behavior:smooth;  
}  
  
body{  
background:#0b0f1a;  
color:white;  
min-height:100vh; /* ✓ FIX */  
overflow-y:auto; /* ✓ FIX */  
}  
  
/* BLOBS */  
.blob{  
position:absolute;  
border-radius:50%;  
filter:blur(120px);  
opacity:0.6;  
animation:move 10s infinite alternate;  
}  
.blob1{width:600px;height:600px;background:#6a11cb;top:-100px;left:-100px;}  
.blob2{width:500px;height:500px;background:#ff6ec4;bottom:-100px;right:-  
100px;}  
  
@keyframes move{  
0%{transform:translate(0,0);}  
}
```

```

100%{transform:translate(150px,100px);}
}

/* MAIN */
.container{
padding:40px;
padding-bottom:120px; /* ✓ EXTRA SCROLL SPACE */
position:relative;
z-index:1;
}

/* CARD */
.card{
background:rgba(255,255,255,0.05);
padding:20px;
border-radius:15px;
margin-bottom:20px;
backdrop-filter:blur(20px);
}

input{
width:100%;
padding:10px;
margin:5px 0;
border:none;
border-radius:10px;
background:rgba(255,255,255,0.1);
color:white;
}

button{
padding:10px;
margin-top:10px;
border:none;
border-radius:10px;
background:linear-gradient(45deg,#ff6ec4,#7873f5);
color:white;
cursor:pointer;
}
</style>
</head>

<body>

<div class="blob blob1"></div>
<div class="blob blob2"></div>

<div class="container">

<h1>Admin Dashboard</h1>

<div class="card">
<h3>Add Resident</h3>
<input id="name" placeholder="Name">
<input id="phone" placeholder="Phone">
<input id="address" placeholder="Address">
<input id="room" placeholder="Room">

```

```

<input id="rent" placeholder="Rent">
<button onclick="add()">Add</button>
</div>

<div class="card">
<h3>Residents</h3>
<button onclick="loadResidents()">Load</button>
<div id="residents"></div>
</div>

<div class="card">
<h3>Complaints</h3>
<button onclick="loadComplaints()">Load</button>
<div id="complaints"></div>
</div>

</div>

<script>

function add(){
fetch("/admin/add",{method:"POST",
headers:{"Content-Type":"application/json"},
body:JSON.stringify({
name:name.value,
phone:phone.value,
address:address.value,
roomNumber:room.value,
rentDue:rent.value
})
}).then(()=>alert("Added"));
}

function loadResidents(){
fetch("/admin/residents")
.then(r=>r.json())
.then(d=>{
residents.innerHTML=d.map(x=>
`${x.name || "-"} | ${x.phone || "-"} | Room ${x.roomNumber || "-"} | 
₹${x.rentDue || 0}`
).join("<br>");
});
}

function loadComplaints(){
fetch("/admin/complaints")
.then(r=>r.json())
.then(d=>{
complaints.innerHTML=d.map(x=>
`${x.residentName || "-"} : ${x.issue || "-"} (${x.status || "Pending"})
<br>Response: ${x.response || ""}
<input id="r${x.id}">
<button onclick="reply(${x.id})">Reply</button>`
).join("<br><br>");
});
}

function reply(id){

```

```

fetch("/admin/complaint/"+id,{
method:"PUT",
headers:{"Content-Type":"application/json"},
body:JSON.stringify({response:document.getElementById("r"+id).value})
});
alert("Replied");
}

```

```

</script>

```

```

</body>

```

```

</html>

```

Index.html

```

<!DOCTYPE html>

```

```

<html>

```

```

<head>

```

```

<title>Smart Hostel</title>

```

```

<link

```

```

href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;700
&display=swap" rel="stylesheet">

```

```

<style>

```

```

*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:'Poppins',sans-serif;
}

```

```

body{
height:100vh;
display:flex;
background:#0b0f1a;
overflow:hidden;
}

```

```

/* BLOBS */
.blob{
position:absolute;
border-radius:50%;
filter:blur(120px);
opacity:0.6;
animation:move 10s infinite alternate ease-in-out;
}

```

```

.blob1{
width:600px;
height:600px;
background:#6a11cb;
top:-100px;
left:-100px;
}

```

```

.blob2{
width:500px;
height:500px;
background:#ff6ec4;
bottom:-100px;
right:-100px;
animation-delay:3s;
}

@keyframes move{
0%{transform:translate(0,0);}
100%{transform:translate(150px,100px);}
}

/* MAIN */
.container{
display:flex;
width:100%;
z-index:1;
}

/* LEFT */
.left{
flex:1;
color:white;
padding:80px;
display:flex;
flex-direction:column;
justify-content:center;
animation:fadeIn 1.5s ease;
}

.left h1{
font-size:50px;
font-weight:700;
margin-bottom:20px;
background:linear-gradient(90deg, #fff, #a8c0ff);
-webkit-background-clip:text;
-webkit-text-fill-color:transparent;
}

.left p{
color:#ccc;
font-size:18px;
max-width:400px;
line-height:1.6;
}

/* RIGHT */
.right{
flex:1;
display:flex;
justify-content:center;
align-items:center;
}

/* CARD */

```

```

.card{
width:360px;
padding:40px;
border-radius:20px;
background:rgba(255,255,255,0.05);
backdrop-filter:blur(25px);
border:1px solid rgba(255,255,255,0.1);
box-shadow:0 10px 40px rgba(0,0,0,0.5);
color:white;
transition:0.4s;
animation:float 4s ease-in-out infinite;
}

/* FLOAT */
@keyframes float{
0%,100%{transform:translateY(0);}
50%{transform:translateY(-10px);}
}

/* HOVER */
.card:hover{
transform:rotateY(5deg) rotateX(5deg) scale(1.02);
}

/* FADE */
@keyframes fadeIn{
from{opacity:0; transform:translateY(20px);}
to{opacity:1; transform:translateY(0);}
}

/* INPUT */
.input-box{
margin:15px 0;
}

input,select{
width:100%;
padding:12px;
border:none;
border-radius:10px;
background:rgba(255,255,255,0.08);
color:white;
outline:none;
transition:0.3s;
}

input:focus, select:focus{
background:rgba(255,255,255,0.15);
transform:scale(1.02);
}

/* BUTTON */
button{
width:100%;
padding:12px;
margin-top:10px;
border:none;
border-radius:10px;

```

```
background:linear-gradient(45deg,#ff6ec4,#7873f5);
color:white;
font-weight:600;
cursor:pointer;
transition:0.3s;
box-shadow:0 0 15px rgba(255,110,196,0.5);
}
```

```
button:hover{
transform:scale(1.07);
box-shadow:0 0 30px rgba(255,110,196,1);
}
```

```
/* FOOTER */
.footer{
text-align:center;
margin-top:15px;
font-size:12px;
color:#aaa;
}
```

```
</style>
</head>
```

```
<body>
```

```
<div class="blob blob1"></div>
<div class="blob blob2"></div>
```

```
<div class="container">
```

```
<div class="left">
<h1>Smart Hostel & PG</h1>
```

```
<p>
A modern intelligent system to manage residents, complaints, and rent
efficiently.
```

```
</p>
</div>
```

```
<div class="right">
<div class="card">
```

```
<h2>Login</h2>
```

```
<div class="input-box">
<select id="role">
<option value="admin">Admin</option>
<option value="resident">Resident</option>
</select>
</div>
```

```
<div class="input-box">
<input type="text" placeholder="Username">
</div>
```

```
<div class="input-box">
<input type="password" placeholder="Password">
```

```

</div>

<button onclick="login()">Login</button>

<div class="footer">© 2026 Smart Hostel</div>

</div>
</div>

</div>

<script>
function login(){
let role=document.getElementById("role").value;

if(role==="admin"){
window.location.href="/admin"; // ✔ CORRECT
}else{
window.location.href="/resident"; // ✔ CORRECT
}
}
}
</script>

</body>
</html>

```

Resident.html

```

<!DOCTYPE html>
<html>
<head>
<title>Resident Dashboard</title>

<link href="https://fonts.googleapis.com/css2?family=Poppins&display=swap"
rel="stylesheet">

<style>

/* Smooth scroll */
html{
scroll-behavior:smooth;
}

body{
background:#0b0f1a;
color:white;
font-family:Poppins;
min-height:100vh; /* ✔ added */
overflow-y:auto; /* ✔ fixed (removed overflow:hidden) */
}

/* BLOBS */
.blob{
position:absolute;
border-radius:50%;
filter:blur(120px);

```

```

opacity:0.6;
animation:move 10s infinite alternate;
}
.blob1{
width:600px;
height:600px;
background:#6a11cb;
top:-100px;
left:-100px;
}
.blob2{
width:500px;
height:500px;
background:#ff6ec4;
bottom:-100px;
right:-100px;
}

@keyframes move{
0%{transform:translate(0,0);}
100%{transform:translate(150px,100px);}
}

/* MAIN */
.container{
padding:40px;
padding-bottom:120px; /* ✓ extra scroll space */
position:relative;
z-index:1;
}

/* CARD */
.card{
background:rgba(255,255,255,0.05);
padding:20px;
border-radius:15px;
margin-bottom:20px;
backdrop-filter:blur(20px);
}

/* INPUT */
input{
width:100%;
padding:10px;
margin-top:10px;
border:none;
border-radius:10px;
background:rgba(255,255,255,0.1);
color:white;
}

/* BUTTON */
button{
padding:10px;
margin-top:10px;
border:none;
border-radius:10px;
background:linear-gradient(45deg,#ff6ec4,#7873f5);

```

```

color:white;
cursor:pointer;
}

</style>
</head>

<body>

<div class="blob blob1"></div>
<div class="blob blob2"></div>

<div class="container">

<h1>Resident Dashboard</h1>

<!-- PROFILE -->
<div class="card">
<h3>Profile</h3>
<button onclick="load()">Load</button>
<p id="profile"></p>
</div>

<!-- RENT -->
<div class="card">
<h3>Upload Rent Proof</h3>
<input type="file" id="file">
<button onclick="upload()">Upload</button>
</div>

<!-- COMPLAINT -->
<div class="card">
<h3>Raise Complaint</h3>
<input id="issue" placeholder="Enter issue">
<button onclick="send()">Submit</button>
</div>

</div>

<script>

let id = 1;

/* LOAD PROFILE */
function load(){
fetch("/resident/"+id)
.then(res=>res.json())
.then(d=>{
if(!d){
profile.innerHTML="No data found";
return;
}

profile.innerHTML =
`Name: ${d.name || "-"}<br>
Phone: ${d.phone || "-"}<br>
Room: ${d.roomNumber || "-"}<br>

```

```

Rent: ₹${d.rentDue || 0}`;
});
}

/* UPLOAD RENT */
function upload(){
let fd = new FormData();
fd.append("file", file.files[0]);

fetch("/resident/upload/"+id,{
method:"POST",
body:fd
}).then(=>alert("Uploaded"));
}

/* SEND COMPLAINT */
function send(){
fetch("/resident/complaint",{
method:"POST",
headers:{"Content-Type":"application/json"},
body:JSON.stringify({
residentName:"Resident",
issue:issue.value
})
}).then(=>alert("Complaint Sent"));
}

</script>

</body>
</html>

```

5. Application.properties

```

spring.datasource.url=jdbc:mysql://localhost:3306/hostel_db
spring.datasource.username=root
spring.datasource.password=root

spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true

server.port=9090

```

6. Pom.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>4.0.3</version>

```

```
<relativePath/>
</parent>

<groupId>com.example</groupId>
<artifactId>SMART_HOSTEL_MANAGEMENT</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>SMART_HOSTEL_MANAGEMENT</name>
<description>Smart Hostel Management System</description>

<properties>
<java.version>17</java.version>
</properties>

<dependencies>

<!-- Spring Boot Web -->
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<!-- Thymeleaf (for HTML templates) -->
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-thymeleaf</artifactId>
</dependency>

<!-- JPA -->
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<!-- MySQL -->
<dependency>
<groupId>com.mysql</groupId>
<artifactId>mysql-connector-j</artifactId>
<scope>runtime</scope>
</dependency>

<!-- DevTools -->
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
<scope>runtime</scope>
<optional>true</optional>
</dependency>

<!-- Lombok -->
<dependency>
<groupId>org.projectlombok</groupId>
<artifactId>lombok</artifactId>
<optional>true</optional>
</dependency>

</dependencies>
```

```
<build>
<plugins>

<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>

</plugins>
</build>

</project>
```